

## Homework 2

程昊一

\*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

**Problem 1 (2A-2).** Prove or give a counterexample: If  $v_1, v_2, v_3, v_4$  span  $V$ , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans  $V$ .

**Proof.** We will prove the statement.

By the definition of spanning, we have that for every  $v \in V$ , there exist scalars  $a_1, a_2, a_3, a_4$  such that

$$v = a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4.$$

Then, we can rewrite the above equation as

$$v = a_1(v_1 - v_2) + (a_1 + a_2)(v_2 - v_3) + (a_1 + a_2 + a_3)(v_3 - v_4) + (a_1 + a_2 + a_3 + a_4)v_4.$$

This means that for every  $v \in V$ , there exist scalars  $b_1, b_2, b_3, b_4$  such that

$$v = b_1(v_1 - v_2) + b_2(v_2 - v_3) + b_3(v_3 - v_4) + b_4v_4.$$

□

**Problem 2 (2A-6).** Show that the list  $(2, 3, 1), (1, -1, 2), (7, 3, c)$  is linearly dependent in  $\mathbb{F}^3$  if and only if  $c = 8$ .

**Proof.** On the one hand, if  $c = 8$ , then we have

$$2(2, 3, 1) + 3(1, -1, 2) - (7, 3, 8) = (0, 0, 0).$$

On the other hand, if the list  $(2, 3, 1), (1, -1, 2), (7, 3, c)$  is linearly dependent in  $\mathbb{F}^3$ , then there exist scalars  $a_1, a_2, a_3$  with  $a_3 \neq 0$  (otherwise the first two vectors would be linearly dependent) such that

$$a_1(2, 3, 1) + a_2(1, -1, 2) + a_3(7, 3, c) = (0, 0, 0).$$

This means that

$$\begin{cases} 2a_1 + a_2 + 7a_3 = 0, \\ 3a_1 - a_2 + 3a_3 = 0, \\ a_1 + 2a_2 + ca_3 = 0. \end{cases}$$

Then

$$0 = -7(2a_1 + a_2 + 7a_3) + 3(3a_1 - a_2 + 3a_3) + 5(a_1 + 2a_2 + ca_3) = (5c - 40)a_3.$$

Since  $a_3 \neq 0$ , we have  $c = 8$ . □

**Problem 3 (2A-7).** (a) Show that if we think of  $\mathbb{C}$  as a vector space over  $\mathbb{R}$ , then the list

$$1 + i, 1 - i$$

is linearly independent.

(b) Show that if we think of  $\mathbb{C}$  as a vector space over  $\mathbb{C}$ , then the list

$$1 + i, 1 - i$$

is linearly dependent.

**Solution.** (a) Suppose that there exist scalars  $a_1, a_2 \in \mathbb{R}$  such that

$$a_1(1 + i) + a_2(1 - i) = 0.$$

By comparing the real and imaginary parts of the above equation, we have

$$a_1 + a_2 = 0, \quad a_1 - a_2 = 0. \quad \Rightarrow \quad a_1 = a_2 = 0.$$

(b) Note that

$$(1 + i) - i(1 - i) = 0.$$

**Problem 4 (2B-4).**

(a) Let  $U$  be the subspace of  $\mathbb{C}^5$  defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 : 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of  $U$ .

(b) Extend the basis in (a) to a basis of  $\mathbb{C}^5$ .

(c) Find a subspace  $W$  of  $\mathbb{C}^5$  such that  $\mathbb{C}^5 = U \oplus W$ .

**Solution.** (a) We parametrize the subspace  $U$  using free variables. From  $6z_1 = z_2$ , we have  $z_2 = 6z_1$ . From  $z_3 + 2z_4 + 3z_5 = 0$ , we have  $z_3 = -2z_4 - 3z_5$ .

Let  $z_1 = s$ ,  $z_4 = t$ ,  $z_5 = u$  be free variables. Then

$$(z_1, z_2, z_3, z_4, z_5) = (s, 6s, -2t - 3u, t, u) = s(1, 6, 0, 0, 0) + t(0, 0, -2, 1, 0) + u(0, 0, -3, 0, 1).$$

Therefore, a basis of  $U$  is

$$\{e_1 := (1, 6, 0, 0, 0), e_2 := (0, 0, -2, 1, 0), e_3 := (0, 0, -3, 0, 1)\}.$$

(b) To extend this to a basis of  $\mathbb{C}^5$ , we add the vectors  $e_4 := (0, 1, 0, 0, 0)$  and  $e_5 := (0, 0, 1, 0, 0)$ . On the one hand, it is easy to verify that the five vectors are linearly independent. On the other hand, for every vector  $(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5$ , we have

$$(z_1, z_2, \dots, z_5) = z_1 e_1 + z_4 e_2 + z_5 e_3 + (z_1 - 6z_2) e_4 + (z_3 + 2z_4 + 3z_5) e_5.$$

(c) Let  $W = \text{span}\{e_4, e_5\}$ . Since  $U = \text{span}\{e_1, e_2, e_3\}$  and notice that  $\{e_1, e_2, \dots, e_5\}$  is a basis of  $\mathbb{C}^5$ , hence  $U \cap W = \{0\}$  and  $U + W = \text{span}\{e_1, e_2, e_3, e_4, e_5\} = \mathbb{C}^5$ . Therefore,  $\mathbb{C}^5 = U \oplus W$ .

**Problem 5 (2B-5).** Suppose  $V$  is finite-dimensional and  $U, W$  are subspaces of  $V$  such that  $V = U + W$ . Prove that there exists a basis of  $V$  consisting of vectors in  $U \cup W$ .

**Proof.** Obviously,  $U \cap W$  is a subspace of  $V$ . Pick a basis  $\mathcal{B}_0$  of  $U \cap W$ ; if  $U \cap W = \{0\}$ , let  $\mathcal{B}_0 = \emptyset$ . Then we can extend  $\mathcal{B}_0$  to a basis  $\mathcal{B}_1$  of  $U$  and a basis  $\mathcal{B}_2$  of  $W$ .

We claim that  $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$  is a basis of  $V$ . On the one hand, for every vector  $v \in V$ , there exist vectors  $u \in U$  and  $w \in W$  such that  $v = u + w$ . Notice that  $u \in \text{span}\mathcal{B}_1$  and  $w \in \text{span}\mathcal{B}_2$ , hence  $v \in \text{span}\mathcal{B}$ , which implies that  $\mathcal{B}$  spans  $V$ .

On the other hand, if the vectors in  $\mathcal{B}$  are linearly dependent, then there exists a linear combination of vectors in  $\mathcal{B}$  whose coefficients are not all zero. Note that  $\mathcal{B} = \mathcal{B}_1 \cup (\mathcal{B}_2 - \mathcal{B}_1)$ , and  $\mathcal{B}_1 \cap (\mathcal{B}_2 - \mathcal{B}_1) = \emptyset$ , then the linear combination can be rewritten as a linear combination of vectors in  $\mathcal{B}_1$  plus a linear combination of vectors in  $\mathcal{B}_2 - \mathcal{B}_1$ . In other words, there exist a vector  $v_1 \in \text{span}\mathcal{B}_1$  and a vector  $v_2 \in \text{span}(\mathcal{B}_2 - \mathcal{B}_1)$  such that  $v_1 + v_2 = 0$  and  $v_1 \neq 0, v_2 \neq 0$ . Then,  $v_2 = -v_1 \in \text{span}\mathcal{B}_1$ , which indicates  $v_2 \in (\text{span}\mathcal{B}_1) \cap (\text{span}(\mathcal{B}_2 - \mathcal{B}_1)) = \{0\}$ , which contradicts the fact that  $v_2 \neq 0$ . Therefore, the vectors in  $\mathcal{B}$  are linearly independent.

Hence,  $\mathcal{B}$  is a basis of  $V$  consisting of vectors in  $U \cup W$ . □

**Problem 6 (2B-11).** Suppose  $V$  is a real vector space. Show that if  $v_1, \dots, v_n$  is a basis of  $V$  (as a real vector space), then  $v_1, \dots, v_n$  is also a basis of the complexification  $V_{\mathbb{C}}$  (as a complex vector space).

**Proof.** First, we show that the list  $v_1, \dots, v_n$  spans  $V_{\mathbb{C}}$ . For every vector  $v \in V_{\mathbb{C}}$ , there exist vectors  $u, w \in V$  such that  $v = u + iw$ . Since  $v_1, \dots, v_n$  is a basis of  $V$ , there exist scalars  $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{R}$  such that

$$u = a_1 v_1 + \dots + a_n v_n, \quad w = b_1 v_1 + \dots + b_n v_n.$$

Then

$$v = u + iw = (a_1 v_1 + \dots + a_n v_n) + i(b_1 v_1 + \dots + b_n v_n) = (a_1 + ib_1)v_1 + \dots + (a_n + ib_n)v_n.$$

This shows that  $v \in \text{span}\{v_1, \dots, v_n\}$ , so the list  $v_1, \dots, v_n$  spans  $V_{\mathbb{C}}$ .

Next, we show that the list  $v_1, \dots, v_n$  is linearly independent in  $V_{\mathbb{C}}$ . Suppose that there exist scalars  $c_1, \dots, c_n \in \mathbb{C}$  such that

$$c_1 v_1 + \dots + c_n v_n = 0.$$

Write  $c_j = a_j + ib_j$  for some  $a_j, b_j \in \mathbb{R}$ . Then

$$(a_1 + ib_1)v_1 + \dots + (a_n + ib_n)v_n = (a_1 v_1 + \dots + a_n v_n) + i(b_1 v_1 + \dots + b_n v_n) = 0.$$

Since  $v_1, \dots, v_n$  is a basis of  $V$ , we have  $a_1 = \dots = a_n = 0$  and  $b_1 = \dots = b_n = 0$ . Therefore,  $c_1 = \dots = c_n = 0$ , which shows that the list  $v_1, \dots, v_n$  is linearly independent in  $V_{\mathbb{C}}$ .

Combining the above two parts, it's proved that  $v_1, \dots, v_n$  is a basis of  $V_{\mathbb{C}}$ .  $\square$

**Problem 7 (2C-6).**

- (a) Let  $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5) = p(6)\}$ . Find a basis of  $U$ .
- (b) Extend the basis in (a) to a basis of  $\mathcal{P}_4(\mathbb{F})$ .
- (c) Find a subspace  $W$  of  $\mathcal{P}_4(\mathbb{F})$  such that  $\mathcal{P}_4(\mathbb{F}) = U \oplus W$ .

**Solution.** (a) Set

$$\begin{cases} e_1(x) := 1, \\ e_2(x) := (x-2)(x-5)(x-6), \\ e_3(x) := x(x-2)(x-5)(x-6). \end{cases}$$

We will show that  $\{e_1, e_2, e_3\}$  is a basis of  $U$ .

On the one hand, for every  $p \in U$ , we have  $p(2) = p(5) = p(6)$ . Let  $c = p(2)$ . Then the polynomial  $p - ce_1$  has roots 2, 5, 6, which means that there exists a linear function  $ax + b$  such that

$$p - ce_1 = (ax + b)(x-2)(x-5)(x-6) \Rightarrow p = ae_3 + be_2 + ce_1.$$

On the other hand, if  $ae_3 + be_2 + ce_1 = 0$  for some scalars  $a, b, c$ , then by comparing the coefficients of  $x^4$ , we have  $a = 0$ . Then by comparing the coefficients of  $x^3$ , we have  $b = 0$ . Finally, by comparing the coefficients of  $x^0$ , we have  $c = 0$ . Therefore, the vectors in  $\{e_1, e_2, e_3\}$  are linearly independent.

Hence,  $\{e_1, e_2, e_3\}$  is a basis of  $U$ .

(b) Set

$$\begin{cases} e_4(x) := (x-2)(x-5), \\ e_5(x) := (x-2)(x-6). \end{cases}$$

We will show that  $\{e_1, e_2, e_3, e_4, e_5\}$  is a basis of  $\mathcal{P}_4(\mathbb{F})$ . On the one hand, for every polynomial  $p \in \mathcal{P}_4(\mathbb{F})$ , it is easy to verify that (by taking  $x = 2, 3, 4, 5, 6$  and use the Fundamental Algebra Theorem)

$$\begin{aligned} p &= p(2)e_1 + \left(\frac{5}{4}p(2) + \frac{2}{3}p(3) - \frac{3}{4}p(4) - p(5) - \frac{1}{6}p(6)\right)e_2 \\ &\quad + \left(-\frac{3}{4}p(2) - \frac{1}{6}p(3) + \frac{1}{4}p(4) + \frac{1}{2}p(5) + \frac{1}{6}p(6)\right)e_3 + (p(6) - p(2))e_4 + (p(5) - p(2))e_5. \end{aligned}$$

On the other hand, we will show that the vectors in  $\{e_1, e_2, e_3, e_4, e_5\}$  are linearly independent. Suppose that there exist scalars  $a_1, a_2, a_3, a_4, a_5$  such that

$$a_1e_1 + a_2e_2 + a_3e_3 + a_4e_4 + a_5e_5 = 0.$$

By comparing the coefficients of  $x^4$ , we have  $a_3 = 0$ . Then by comparing the coefficients of  $x^3$ , we have  $a_2 = 0$ . Then by comparing the coefficients of  $x^2$  and  $x$ , we have  $a_4 = a_5 = 0$ . Finally, by comparing the coefficients of  $x^0$ , we have  $a_1 = 0$ . Therefore,  $a_1 = a_2 = a_3 = a_4 = a_5 = 0$ , which shows that the vectors in  $\{e_1, e_2, e_3, e_4, e_5\}$  are linearly independent.

(c) Let  $W = \text{span}\{e_4, e_5\}$ . Since  $U = \text{span}\{e_1, e_2, e_3\}$  and notice that  $\{e_1, e_2, e_3, e_4, e_5\}$  is a basis of  $\mathcal{P}_4(\mathbb{F})$ , hence  $U \cap W = \{0\}$  and  $U + W = \text{span}\{e_1, e_2, e_3, e_4, e_5\} = \mathcal{P}_4(\mathbb{F})$ . Therefore,  $\mathcal{P}_4(\mathbb{F}) = U \oplus W$ .

**Problem 8 (2C-8).** Suppose  $v_1, v_2, \dots, v_m$  is linearly independent and  $w \in V$ . Prove that

$$\dim \text{span}\{v_1 + w, v_2 + w, \dots, v_m + w\} \geq m - 1.$$

**Proof.** Denote  $v_i + w$  as  $v'_i$  and  $U = \text{span}\{v_1 + w, \dots, v_m + w\} = U$ . We will prove the problem by two steps.

(i) First, we show that the  $m - 1$  vectors  $v_1 - v_m, v_2 - v_m, \dots, v_{m-1} - v_m$  are linearly independent. Suppose that there exist not-all-zero scalars  $a_1, a_2, \dots, a_{m-1}$  such that

$$a_1(v_1 - v_m) + a_2(v_2 - v_m) + \dots + a_{m-1}(v_{m-1} - v_m) = 0.$$

We can rewrite the above equation as

$$a_1v_1 + a_2v_2 + \dots + a_{m-1}v_{m-1} - (a_1 + a_2 + \dots + a_{m-1})v_m = 0.$$

Since  $v_1, v_2, \dots, v_m$  are linearly independent, we have  $a_1 = a_2 = \dots = a_{m-1} = 0$ , which contradicts the fact that the scalars are not all zero. Therefore, the  $m - 1$  vectors  $v_1 - v_m, v_2 - v_m, \dots, v_{m-1} - v_m$  are linearly independent.

(ii) Next, Note that

$$\begin{aligned} U &\supset \left\{ \sum_{i=1}^m a_i v_i : a_i \in \mathbb{R}, \sum_{i=1}^m a_i = 0 \right\} \\ &= \left\{ \sum_{i=1}^{m-1} a_i v_i - \left( \sum_{i=1}^{m-1} a_i \right) v_m : a_i \in \mathbb{R}, \sum_{i=1}^m a_i = 0 \right\} \\ &= \left\{ \sum_{i=1}^{m-1} a_i (v_i - v_m) : a_i \in \mathbb{R} \right\} \\ &= \text{span}\{v_1 - v_m, \dots, v_{m-1} - v_m\}. \end{aligned}$$

Then By the result in (i), we have

$$\dim U \geq \dim \text{span}\{v_1 - v_m, \dots, v_{m-1} - v_m\} = m - 1.$$

□

**Problem 9** (2C-16). Suppose  $V$  is finite-dimensional and  $U$  is a subspace of  $V$  with  $U \neq V$ . Let  $n = \dim V$  and  $m = \dim U$ . Prove that there exist  $n - m$  subspaces of  $V$ , each of dimension  $n - 1$ , whose intersection equals  $U$ .

**Proof.** Take a basis  $u_1, \dots, u_m$  of  $U$  and extend it to a basis  $u_1, \dots, u_m, v_1, \dots, v_{n-m}$  of  $V$ . For each  $i = 1, \dots, n - m$ , let

$$W_i = \text{span}\{u_1, \dots, u_m, v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_{n-m}\}.$$

Then  $W_i$  is a subspace of  $V$  with dimension  $n - 1$  (since it is spanned by  $n - 1$  linear independent vectors). We will show that the intersection of these subspaces equals  $U$ .

On the one hand, since  $U \subset W_i$  for each  $i$ , we have  $U \subset \bigcap_{i=1}^{n-m} W_i$ .

On the other hand, if  $v \in \bigcap_{i=1}^{n-m} W_i$ , then  $v \in W_i$  for each  $i$ . Writing  $v$  as a linear combination of the basis vectors, we have

$$v = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_{n-m} v_{n-m}.$$

For each  $1 \leq i \leq n - m$ ,  $v - b_i v_i$  is a linear combination of  $u_1, \dots, u_m, v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_{n-m}$  which are all in  $V$ , but  $v_i \notin W_i$ , consequently we have  $b_i = 0$ . Therefore,  $v = a_1 u_1 + \dots + a_m u_m \in U$ . This shows that  $\bigcap_{i=1}^{n-m} W_i \subset U$ .

Combining the above two parts, we have  $\bigcap_{i=1}^{n-m} W_i = U$ .  $\square$

**Problem 10** (3A-4). Suppose  $T \in \mathcal{L}(V, W)$  and  $v_1, \dots, v_m$  is a list of vectors in  $V$  such that  $Tv_1, \dots, Tv_m$  is a linearly independent list in  $W$ . Prove that  $v_1, \dots, v_m$  is linearly independent.

**Proof.** Suppose that there exist scalars  $a_1, \dots, a_m$  such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

Applying  $T$  to both sides of the above equation, we have

$$a_1 Tv_1 + \dots + a_m Tv_m = 0.$$

Since  $Tv_1, \dots, Tv_m$  is a linearly independent list in  $W$ , we have  $a_1 = \dots = a_m = 0$ . Therefore,  $v_1, \dots, v_m$  is linearly independent.  $\square$

**Problem 11** (3A-11). Suppose  $V$  is finite-dimensional and  $T \in \mathcal{L}(V)$ . Prove that  $T$  is a scalar multiple of the identity if and only if  $ST = TS$  for every  $S \in \mathcal{L}(V)$ .

**Proof.** On the one hand, if  $T$  is a scalar multiple of the identity, then  $T = \lambda I$  for some scalar  $\lambda$ . For every  $S \in \mathcal{L}(V)$ , we have

$$ST = S(\lambda I) = \lambda S = (\lambda I)S = TS.$$

On the other hand, if  $ST = TS$  for every  $S \in \mathcal{L}(V)$ , then we will show that  $T$  is a scalar multiple of the identity. Pick a basis  $v_1, \dots, v_n$  of  $V$ . For each  $1 \leq i \leq n$ , define the linear map  $S_i$  as follow: for arbitraty  $v := \sum_{i=1}^n a_i v_i \in V$ ,  $S_i(v) = a_i v_i$ . For  $i, j \in \{1, 2, \dots, n\}$ , set

$$T(v_j) = c_{j1}v_1 + c_{j2}v_2 + \dots + c_{jn}v_n.$$

Then

$$S_i T(v_j) = S_i(c_{j1}v_1 + c_{j2}v_2 + \dots + c_{jn}v_n) = c_{ji}v_i,$$

and

$$T S_i(v_j) = \delta_{ij} T(v_j) = a_j T(v_j) = \delta_{ij}(c_{j1}v_1 + c_{j2}v_2 + \dots + c_{jn}v_n).$$

Since  $S_i T = T S_i$ , by comparing coefficients, we have  $c_{ji} = 0$  for every  $i \neq j$ . Therefore, for every  $1 \leq j \leq n$ , there exists a scalar  $c_j$  such that

$$T(v_j) = c_j v_j.$$

Next, define  $S'_i$  for  $2 \leq i \leq n$  as follow: for arbitraty  $v := \sum_{i=1}^n a_i v_i \in V$ ,  $S'_i(v) = a_i v_i$ . Then

$$S'_i T(v_1) = S'_i(c_1 v_1) = c_1 v_i, \quad T S'_i(v_1) = T(v_i) = c_i v_i.$$

We immedieately have  $c_1 = c_i$  for every  $2 \leq i \leq n$ , which indicates

$$T(v) = c_1 v.$$

□

**Problem 12 (3A-13).** Suppose  $V$  is finite-dimensional. Prove that every linear map on a subspace of  $V$  can be extended to a linear map on  $V$ . In other words, show that if  $U$  is a subspace of  $V$  and  $S \in \mathcal{L}(U, W)$ , then there exists  $T \in \mathcal{L}(V, W)$  such that  $Tu = Su$  for all  $u \in U$ .

**Proof.** Pick a basis  $u_1, \dots, u_m$  of  $U$  and extend it to a basis  $u_1, \dots, u_m, v_1, \dots, v_{n-m}$  of  $V$ . Denote  $\text{span}\{v_1, \dots, v_{n-m}\}$  by  $U'$ , then  $V = U \oplus U'$ . For arbitraty  $v \in V$ , we can write  $v$  as

$$v = u + u', \quad u \in U, \quad u' \in U'.$$

Define  $T \in \mathcal{L}(V, W)$  as  $T(v) = S(u)$ . Then it's easy to verify that for all  $v \in U$ ,  $T(v) = S(v)$ . □

**Problem 13 (3B-4).** Show that  $T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \ker T > 2$  is not a subspace of  $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$ .

**Proof.** We will show that the set  $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \ker T > 2\}$  is not closed under addition.

Define the linear map  $S \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$  as

$$S(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, 0).$$

Then  $S$  is a linear map with  $\ker S = \{(0, 0, x_3, x_4, x_5) : x_3, x_4, x_5 \in \mathbb{R}\}$  and  $\dim \ker S = 3$ . Then, define the linear map  $T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$  as

$$T(x_1, x_2, x_3, x_4, x_5) = (0, 0, x_4, x_5).$$

Then it's clear that  $\dim \ker T = 3$ .

However,  $S + T$  is a linear map defined as

$$(S + T)(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, x_4, x_5).$$

Then  $\ker(S + T) = \{(0, 0, x_3, 0, 0) : x_3 \in \mathbb{R}\}$  and  $\dim \ker(S + T) = 1$ . Therefore,  $S + T \notin \{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \ker T > 2\}$ , which shows that the set  $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) : \dim \ker T > 2\}$  is not closed under addition.  $\square$

**Problem 14 (3B-22).** Suppose  $U$  and  $V$  are finite-dimensional vector spaces and  $S \in \mathcal{L}(V, W)$  and  $T \in \mathcal{L}(U, V)$ . Prove that

$$\dim \ker ST \leq \dim \ker S + \dim \ker T.$$

**Proof.** Let

$$\dim \ker S = s, \quad \dim \ker T = t, \quad \dim \ker ST = d.$$

It's clear that  $\ker T \subset \ker ST$ . Select a basis of  $\ker T$ , denoted as  $\{v_1, v_2, \dots, v_t\}$  and extend it to a basis of  $\ker ST$ , denoted as  $\{v_1, v_2, \dots, v_t, u_1, \dots, u_{d-t}\}$ .

First, we claim that  $\{Tu_1, Tu_2, \dots, Tu_{d-t}\}$  is a linearly independent list in  $V$ . Suppose that there exist scalars  $a_1, a_2, \dots, a_{d-t}$  such that

$$a_1 Tu_1 + a_2 Tu_2 + \dots + a_{d-t} Tu_{d-t} = T(u) = 0,$$

where  $u := \sum_{i=1}^{d-t} a_i u_i$ . Then  $u \in \ker T = \text{span}\{v_i : 1 \leq i \leq t\}$ , which contradicts the fact that  $\{v_1, v_2, \dots, v_t, u_1, \dots, u_{d-t}\}$  is a basis of  $\ker ST$ . Therefore,  $\{Tu_1, Tu_2, \dots, Tu_{d-t}\}$  is a linearly independent list in  $V$ .

Next, we claim that  $\{Tu_1, Tu_2, \dots, Tu_{d-t}\} \subset \ker S$ . For each  $1 \leq i \leq d - t$ , since  $u_i \in \ker ST$ , we have  $S(Tu_i) = 0$ , which means that  $Tu_i \in \ker S$ . Therefore,  $\{Tu_1, Tu_2, \dots, Tu_{d-t}\} \subset \ker S$ .

Since  $\{Tu_1, Tu_2, \dots, Tu_{d-t}\}$  is a linearly independent list in  $\ker S$ , we have  $d - t \leq s$ . Therefore,  $d \leq s + t$ .  $\square$

**Problem 15 (3B-25).** Suppose that  $W$  is finite-dimensional and  $S, T \in \mathcal{L}(V, W)$ . Prove that  $\ker S \subset \ker T$  if and only if there exists  $E \in \mathcal{L}(W)$  such that  $T = ES$ .

**Proof.** On the one hand, if there exists  $E \in \mathcal{L}(W)$  such that  $T = ES$ , then for every  $v \in \ker S$ , we have

$$T(v) = ES(v) = E(0) = 0,$$

which shows that  $v \in \ker T$ . Therefore,  $\ker S \subset \ker T$ .

On the other hand, if  $\ker S \subset \ker T \subset V$ , then by the theorem of basis extension, we can find 3 sets of vectors

$$\mathcal{A} := \{u_1, \dots, u_a\} \quad \mathcal{B} := \{v_1, \dots, v_b\} \quad \mathcal{C} := \{w_1, \dots, w_c\},$$



such that  $\mathcal{A}$  is a basis of  $\ker S$ ,  $\mathcal{A} \cup \mathcal{B}$  is a basis of  $\ker T$ , and  $\mathcal{A} \cup \mathcal{B} \cup \mathcal{C}$  is a basis of  $V$ . We can prove (by the same method in the previous exercise) that  $\{S(x) : x \in \mathcal{B} \cup \mathcal{C}\}$  is linearly independent.

Denote

$$\mathcal{B}' := \{S(v_1), S(v_2), \dots, S(v_b)\}, \quad \mathcal{C}' := \{S(w_1), S(w_2), \dots, S(w_c)\}.$$

Then  $\mathcal{B}' \cup \mathcal{C}'$  is linearly independent, which implies we can extend it to a basis of  $V$ , denoted as  $\mathcal{A}' \cup \mathcal{B}' \cup \mathcal{C}'$ .

Then we can define linear map  $E \in \mathcal{L}(W)$  by defining  $E$  on this basis as follow:

$$\begin{cases} E(a') = 0, & \forall a' \in \mathcal{A}'; \\ E(S(v_i)) = T(v_i) = 0, & \forall v_i \in \mathcal{B}; \\ E(S(w_i)) = T(w_i), & \forall w_i \in \mathcal{C}. \end{cases}$$

Let's verify that  $T = ES$ . We only need to verify the two linear maps are the same on the basis  $\mathcal{A} \cup \mathcal{B} \cup \mathcal{C}$ .

- For every  $a \in \mathcal{A}$ ,  $ES(a) = E(0) = 0 = T(a)$  since  $a \in \ker S \subset \ker T$ .
- For every  $x \in \mathcal{B} \cup \mathcal{C}$ , we have  $ES(x) = T(x)$  by the definition of  $E$ .

Therefore, the statement is proved.  $\square$

**Problem 16 (3B-27).** Suppose  $P \in \mathcal{L}(V)$  such that  $P^2 = P$ . Prove that  $V = \ker P \oplus \operatorname{im} P$ .

**Proof.** First, we will show that  $\ker P \cap \operatorname{im} P = \{0\}$ . If  $v \in \ker P \cap \operatorname{im} P$ , then there exists  $u \in V$  such that  $v = Pu$ . Since  $v \in \ker P$ , we have

$$0 = Pv = P(Pu) = P^2u = Pu = v.$$

Therefore,  $v = 0$ , which shows that  $\ker P \cap \operatorname{im} P = \{0\}$ .

Next, we will show that  $V = \ker P + \operatorname{im} P$ . For every  $v \in V$ , we can write

$$v = (v - Pv) + Pv.$$

Note that  $Pv \in \operatorname{im} P$ . Moreover,

$$P(v - Pv) = Pv - P^2v = Pv - Pv = 0 \Rightarrow v - Pv \in \ker P.$$

Therefore,  $v \in \ker P + \operatorname{im} P$ . This shows that  $V = \ker P + \operatorname{im} P$ . Combining the above two parts, we have  $V = \ker P \oplus \operatorname{im} P$ .  $\square$